

FRTT350J Rear Helper Spring Kit Installation Manual

Model: FRTT350J (Rear Suspension Package)

Please read these instructions carefully before commencing installation. Ensure all safety precautions are observed, and the vehicle is properly supported. This document provides detailed procedures for the correct installation of the rear helper spring kit.

1. Parts List

Before beginning, verify that all components listed in the table below are present in your kit.

Ref #	Description / Specifications	Qty
1	Weldment Bracket (Left-Hand & Right-Hand)	2
2	Hollow Rubber Spring	2
3	Hex Cap Screw (M10 x 40 x 1.5, Grade 8.8, Zinc Plated)	2
4	Hex Cap Screw (M10 x 65 x 1.5, Grade 8.8, Zinc Plated)	2
5	Shim Plate (1-3/8" Length x 1/8" Width x 1/32" Thickness, 5/8" Slot)	4
6	Rubber Spacer (4" Outer Diameter x 1" Thickness, with Bushing)	2
7	Fender Washer (3/8" Inner Diameter x 1-1/2" Outer Diameter, Grade 2, Zinc Plated)	2

2. Installation Instructions

1. Prepare Vehicle and Remove OEM Bump Stop: Safely elevate the vehicle using an appropriate lift or jack stands, ensuring it is stable and secure. Locate the existing original equipment manufacturer (OEM) bump stop assembly on the rear suspension. Carefully remove the OEM bump stop assembly, retaining any fasteners for potential re-use.

2. Assemble Helper Spring Unit: Take one hollow rubber spring (Ref 2) and one weldment bracket (Ref 1). Position the rubber spring onto the bracket. Secure this assembly using one M10 x 40 hex cap screw (Ref 3) and one fender washer (Ref 7) through the designated holes. Ensure the bracket is oriented correctly with the 'Front' and 'Wheel Side' markings as indicated in the reference diagram. Repeat this assembly process for the opposite side of the vehicle.

3. Install Helper Spring Assembly: Mount the assembled helper spring unit to the vehicle's frame utilizing the existing OEM mounting holes. Use the retained OEM fasteners and any necessary shim plates (Ref 5) to achieve proper alignment and fitment. If the OEM vehicle features a peg-style casting at the bump stop location, ensure the hollow rubber spring is positioned to contact this peg; do not remove the OEM peg.

4. Adjust Ride Height (Optional): To achieve a desired ride height or to introduce pre-load, an optional rubber spacer (Ref 6) can be inserted between the hollow rubber spring and the weldment bracket. If utilizing the spacer, secure it in place using the longer M10 x 65 hex cap screw (Ref 4) in conjunction with the fender washer (Ref 7).

5. Final Inspection: Verify that all fasteners are tightened securely to the vehicle manufacturer's recommended torque specifications. Conduct a thorough inspection to ensure no brake lines, brake cables, or other vehicle components interfere with the helper spring assembly, particularly when the suspension is articulated or under load.

Installation Notes:

- Verify that brake lines or cables do not make contact with the helper spring assembly during suspension travel.
- If the original equipment manufacturer (OEM) vehicle is equipped with a peg-style casting at the bump stop location, the helper spring is designed to make contact with this peg. Do not remove the OEM peg.
- The use of the rubber spacer (Ref 6) and longer M10 x 65 hex cap screw (Ref 4) allows for adjustment of the vehicle's ride height or pre-load.

3. Clearance Instructions & Fitting Guidance

- Clearance Gap: For light-duty trucks and vans, an initial gap of 1/2 inch to 1-1/2 inches should be maintained between the helper spring and its contact point when the vehicle is unloaded.
- Load Effects: The presence of a partial load or additional vehicle accessories will reduce or may eliminate this initial clearance gap.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended gross axle weight rating (GAWR).
- Inspect for potential interference between the helper spring and other vehicle components. Under load, the helper spring's diameter will expand, requiring adequate surrounding clearance.

Note: Illustrations are for reference only. If no specific reference image is provided, please install according to the standard mechanical guidelines.